

Alpha Model Reference

VSEPR-SIM 3.0.0 — Condensed Booklet

VSEPR-SIM Project

April 14, 2026

1 Overview and Production Architecture

The Alpha Model predicts scalar isotropic polarizability α ($\text{\AA}^3$) for any element by atomic number Z . Production code lives in `atomistic/models/alpha_model.hpp`.

Dual-path architecture:

Path	Range	Error
Empirical table	$Z = 1\text{--}118$	0% (by construction)
Generative model	$Z > 118$	$\sim 15\text{--}17\%$ global RMS (historical best)

For $Z \leq 118$, α is returned directly from `ALPHA_EMPIRICAL[Z-1]`, a 118-element compile-time table. For $Z > 118$, the generative fallback fires.

2 Empirical Table

The table contains experimentally determined and high-level ab initio polarizabilities for every known element:

- **$Z = 1\text{--}54$** : gas-phase experimental values from Miller, *JACS* **112**, 8533 (1990).
- **$Z = 21\text{--}118$** : DHF/CCSD(T) recommended values from Schwerdtfeger & Nagle, *Mol. Phys.* **117**, 1200 (2019).

For overlapping elements ($Z = 21\text{--}54$), experimental values are preferred where available; computational values fill the heavy-element range. The table is the sole prediction path in production and carries 0% model error by construction.

3 Generative Fallback Model

The generative model exists only for hypothetical super-heavy elements ($Z > 118$). During development it was also trained against the $Z = 1\text{--}118$ reference data to validate its functional form. It never exceeded 15% global RMS accuracy on that training set.

3.1 Model Equation

$$\alpha(Z) = r_{\text{eff}}(Z)^3 \cdot g_{\text{block}}(Z) \cdot g_{\text{period}}(Z) \cdot g_f(n_f(Z)) \cdot g_{\text{bind}}(Z) + \Delta_f(Z) \quad (1)$$

where each gate factor is:

$$r_{\text{eff}} = r_{\text{cov}} (1 + c_{\text{rel}} Z^2) (1 - \beta_f n_f / 14) \quad (2)$$

$$g_{\text{block}} = c_{\text{block}}[b(Z)], \quad b \in \{s, p, d, f\} = \{0, 1, 2, 3\} \quad (3)$$

$$g_{\text{period}} = k_{\text{period}}[p(Z) - 1], \quad p \in 1 \dots 7 \quad (4)$$

$$g_f(n_f) = 1 + a_{f1} \frac{n_f}{14} + a_{f2} \exp\left[-\left(\frac{n_f - 7}{\sigma_{f1}}\right)^2\right] + a_{f3} \exp\left[-\left(\frac{n_f - 14}{\sigma_{f2}}\right)^2\right] \quad (5)$$

$$g_{\text{bind}} = \frac{1}{1 + b_{\text{bind}} I_1(Z)} \quad (6)$$

The additive blob correction Δ_f fires only for f-block elements:

$$\Delta_f(Z) = b_{f,\text{lin}} n_f + b_{f,\text{half}} \delta_{\text{half}}(Z) + b_{f,\text{full}} \delta_{\text{full}}(Z) \quad (7)$$

3.2 Physical Interpretation of Gate Factors

r_{eff} (**Eq. ??**) Covalent radius with relativistic contraction ($c_{\text{rel}} < 0$) and lanthanide/actinide contraction (β_f): 4f/5f electrons are poor shielders, so Z_{eff} rises with filling.

g_f (**Eq. ??**) Shell-closure gate. Two Gaussians at half-shell ($n_f = 7$: Eu, Am) and full-shell ($n_f = 14$: Yb, No) capture subshell-stabilization anomalies.

g_{bind} (**Eq. ??**) First ionization energy as a stiffness measure: tightly bound electrons polarize less.

Δ_f (**Eq. ??**) Additive correction for electronic-configuration discontinuities that the smooth r_{cov}^3 basis cannot represent.

3.3 Fitted Parameters (v2.6.2 snapshot)

15 free parameters in total (7 period + 4 block + 4 model).

Symbol	Role	Value
$k_{\text{period},1-7}$	period scaling	20.82, 5.38, 5.12, 3.96, 3.73, 2.92, 2.67
$c_{\text{block},s/p/d/f}$	block correction	1.46, 0.97, 1.61, 1.75
c_{rel}	relativistic contraction	0.0 (cold)
β_f	f-shielding contraction	0.0 (cold)
b_{bind}	ionization stiffness	0.0 (cold)
q_{drude}	Drude coupling charge	1.0 e

Parameters marked ‘cold’ were never moved from their initial values during fitting—the optimizer found no gradient signal through them at the resolution available.

4 Performance Summary

4.1 Best Achieved Accuracy

Some element blocks approached good tolerance while others remained stubbornly above the 15% floor.

Block / Subset	Best RMS (%)
Halogens (F, Cl, Br, I, At)	14.2
Period 6 (Cs–Rn)	14.1
s-block (alkali + alkaline earth)	14.8
p-block (main-group non-metals)	15.0
d-block (transition metals)	15.5
f-block (lanthanides + actinides)	17.3
Global	15.3

4.2 Version Progression

Version	Global RMS	Key Change
v2.0	34%	r_{cov}^3 only
v2.2	27%	+ period/block gates
v2.4	21%	+ ionization binding
v2.6	17%	+ f-block Gaussians
v2.6.2	15.3%	+ additive blob
v2.8	15.1%	+ relativistic + shielding (marginal)

The model plateaued at $\sim 15\%$ between v2.6 and v2.8. Additional parameters produced negligible improvement, confirming that the functional form had reached its representational ceiling.

5 Root-Cause Analysis of the 15–17% Floor

Four structural causes were identified:

1. **Descriptor degeneracy in the f-block.** The descriptor set (cov-radius, period, block, f-electron count, I_1) maps multiple chemically distinct elements to nearly identical descriptor vectors. Irreducible degeneracy of $\sim 6\text{--}8\%$ within the f-block.
2. **Anomalous elements.** Pd ($Z = 46$, $4d^{10}$), Eu/Am half-shell maxima, Yb/No full-shell drops, Lr ($Z = 103$, post-actinide spike). These elements have electronic configurations that break the smooth periodic trends the model assumes.
3. **Single-radius basis limitation.** The model uses r_{cov}^3 as its sole volumetric basis. Polarizability depends on the outer electron cloud, which is not strictly proportional to covalent radius across all blocks.
4. **Absence of many-body or correlation physics.** Scalar polarizability in heavy atoms is influenced by core-valence correlation and spin-orbit effects that no descriptor-free analytical model can capture without explicit electronic structure calculation.

6 Decision: Empirical Table as Production Path

Given the irreducible 15% floor, a deliberate architectural decision was made in v2.8.X:

The empirical table is the primary and sole production path for $Z = 1\text{--}118$. The generative model is retained only as a fallback for hypothetical super-heavy elements ($Z > 118$) where no reference data exists.

This decision eliminated 100% of prediction error for all known elements while preserving the generative path for exploratory use.

7 Anti-Pattern Log

Anti-Pattern	Lesson
Over-fitting f-block	Adding more f-block Gaussians beyond half/full-shell produced oscillatory fits with worse out-of-sample error.
Polynomial Z terms	Replacing r_{cov}^3 with Z^n polynomials destroyed physical interpretability for zero RMS gain.
Per-element offsets	Adding per-element additive constants collapsed the model to a table with extra steps.
Aggressive regularization	L2 penalty strong enough to prevent over-fitting also prevented the optimizer from finding the 15% basin.

8 SCF Polarization Integration

The Alpha Model feeds the Drude/SCF polarization solver through:

$$k_D(Z) = \frac{q_D^2}{\alpha(Z)} \quad (8)$$

where q_D is the Drude particle charge (default 1.0 e). This couples static polarizability to the iterative self-consistent-field loop via Thole-style damped dipole interactions.

9 File Index

File	Purpose
atomistic/models/alpha_model.hpp	Production prediction (table + fallback)
atomistic/models/atomic_descriptors.hpp	Descriptor layer
atomistic/models/atomic_descriptors_ext.hpp	Extended descriptors
data/polarizability_ref.csv	Reference dataset
config/alpha_model_params.json	Generative fallback parameters
tools/fit_alpha_model.cpp	Diagnostic offline fitter
tools/auto_fit_alpha.cpp	Automated fitting harness
tests/test_alpha_model.cpp	Invariant tests (T1–T4)

References

1. Miller, T.M. *JACS* **112**, 8533 (1990).
2. Schwerdtfeger, P.; Nagle, J.K. *Mol. Phys.* **117**, 1200 (2019).
3. Thole, B.T. *Chem. Phys.* **59**, 341 (1981).
4. Lamoureux, G.; Roux, B. *J. Chem. Phys.* **119**, 3025 (2003).
5. NIST Atomic Spectra Database, <https://www.nist.gov/pml/atomic-spectra-database>.